fun-miRBench Benchmark Report

Run setting	Value
Experiments evaluated	21
Experiments skipped	0
Predictors evaluated	5
Responsive genes	oriented shrunken log2FC > 0; repression FSOS s-value < 0.05 against d = 0.5
No-response genes	whole shrunken 95% interval of the oriented change below d = 0.5

Evaluation design

- Every predictor is scored on the same 19,226 genes; unscored genes rank last.

- Unresolved genes are left out of the AUROC.

- AUROC on 16 of 21 experiments (at least 30 responsive and 30 no-response genes).

- Spearman on all 21 experiments.

Key results

- Highest mean AUROC: miRBind2-3UTR (0.637).

- Highest mean Spearman: miRBind2-3UTR (0.095).

- Highest responsive pair coverage: miRBind2-3UTR (99.9%).

- Lowest responsive pair coverage: miRDB (21.6%).

Output tables

- per_comparison.tsv: every metric, for each experiment and tool.

- summary.tsv: means and 95% intervals, on all experiments and on unflagged ones.

- coverage.tsv: genes and pairs covered, and experiments retained, per tool.

- recovery.tsv: recall per experiment and tool, at each prediction budget.

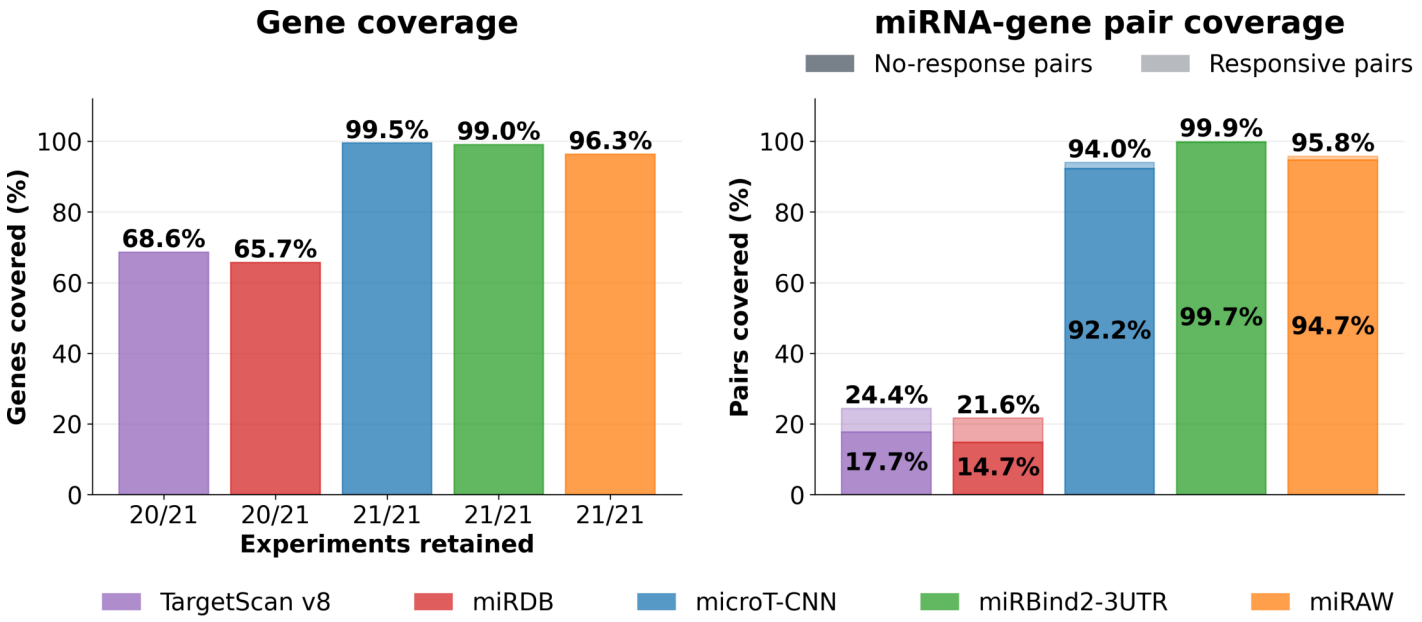
- precision_at_recall.tsv: precision and false positives at 50% recall.

Coverage

- Gene coverage: share of measured genes the tool scores in at least one experiment.

- Experiments retained: experiments where it scores responsive and no-response genes.

- Pair coverage: share of responsive and of no-response miRNA-gene pairs it scores.

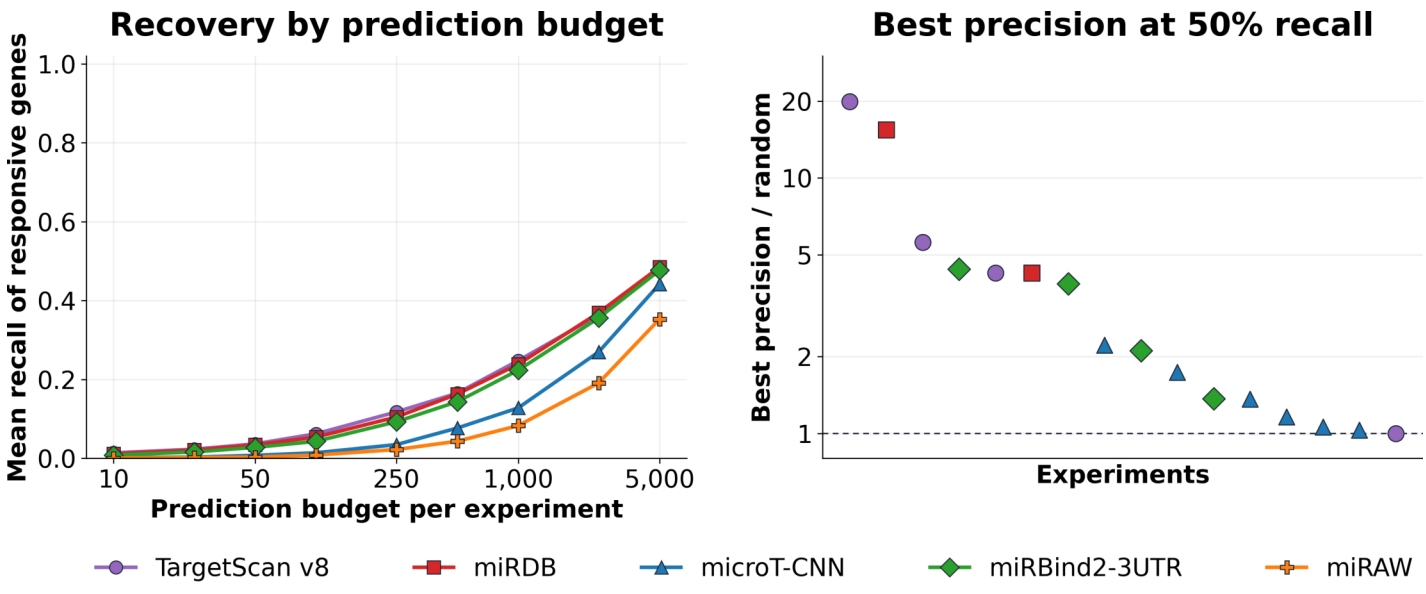


Recovery

- Recall: share of responsive genes in the top genes of the full ranking, mean over experiments.

- Best precision / random: best tool's precision at 50% recall over responsive share.

- Precision counts responsive and no-response genes only.



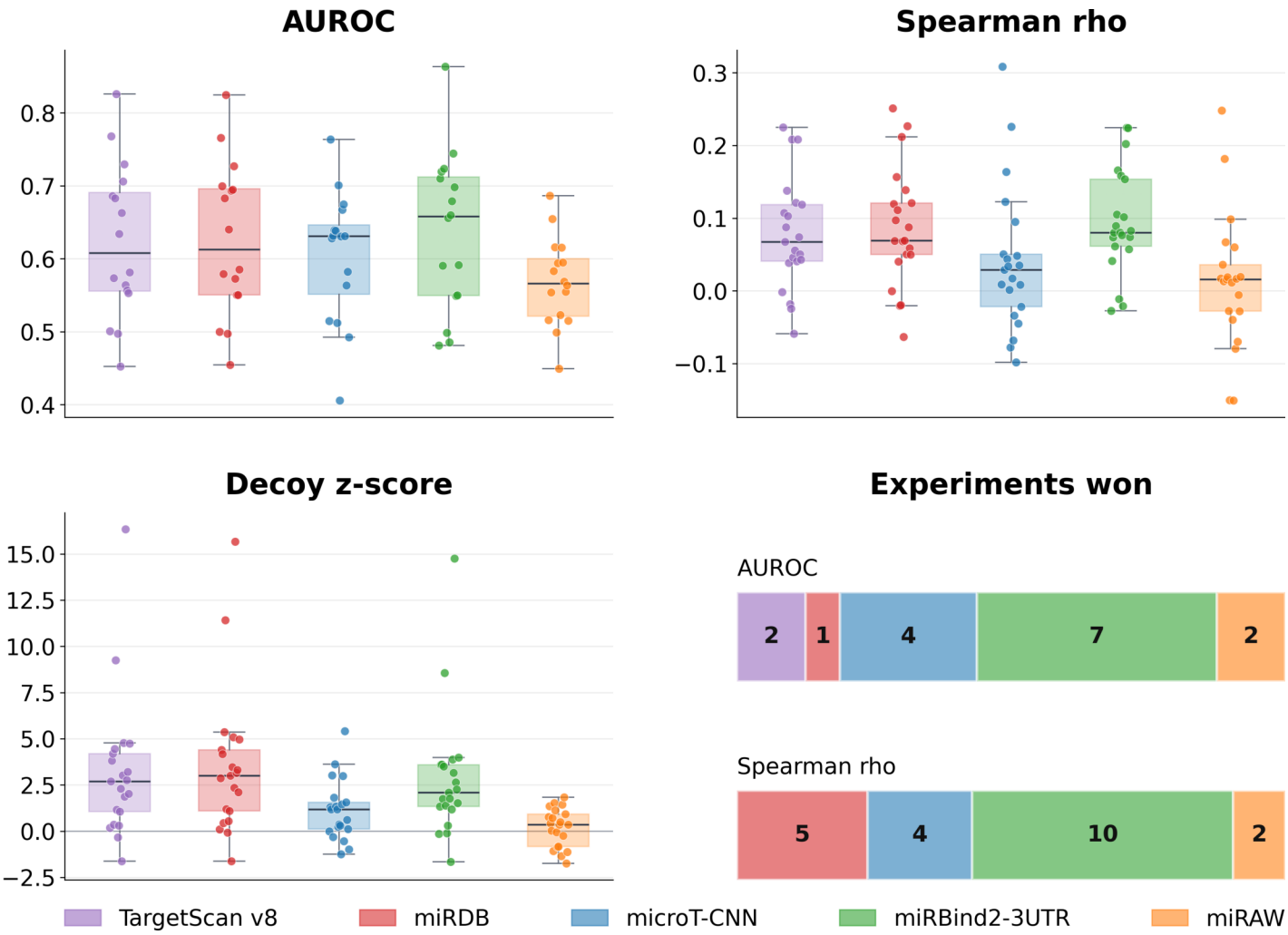
Performance

- AUROC: chance that a responsive gene scores above a no-response gene; 0.5 is chance.

- Spearman rho: agreement of the score's ranking with the shrunken log2FC; 0 is none.

- Decoy z-score: Spearman for the tested miRNA against different-seed miRNAs, in SDs.

- Experiments won: experiments where the tool has the highest AUROC or Spearman rho.



Coverage

Predictor	Gene coverage	Experiments retained	Responsive pairs	No-response pairs
TargetScan v8	68.6%	20/21	24.4%	17.7%
miRDB	65.7%	20/21	21.6%	14.7%
microT-CNN	99.5%	21/21	94.0%	92.2%
miRBind2-3UTR	99.0%	21/21	99.9%	99.7%
miRAW	96.3%	21/21	95.8%	94.7%

Performance

Predictor	Mean AUROC	Mean Spearman	Mean decoy z-score	Won, AUROC	Won, Spearman
TargetScan v8	0.623	0.078	3.169	2	0
miRDB	0.626	0.087	3.478	1	5
microT-CNN	0.605	0.040	1.116	4	4
miRBind2-3UTR	0.637	0.095	2.827	7	10
miRAW	0.568	0.012	0.194	2	2

Recovery

Predictor	Mean recall at 5,000 genes	Experiments won at 50% recall
TargetScan v8	0.482	4
miRDB	0.485	2
microT-CNN	0.442	6
miRBind2-3UTR	0.477	4
miRAW	0.353	0

Experiment inventory

Experiment	Experiment ID
hsa-miR-574-5p · MGC-803	GSE112859_OE_miR_574_5p
hsa-miR-101-3p · HepG2	GSE115646_OE_miR-101-3p_HepG2
hsa-miR-133a-3p · HepG2	GSE115646_OE_miR-133a-3p_HepG2
hsa-miR-140-3p · HepG2	GSE115646_OE_miR-140-3p_HepG2
hsa-miR-140-5p · HepG2	GSE115646_OE_miR-140-5p_HepG2
hsa-miR-18a-5p · HepG2	GSE115646_OE_miR-18a-5p_HepG2
hsa-miR-211-5p · HepG2	GSE115646_OE_miR-211-5p_HepG2
hsa-miR-150-5p · LX-2	GSE124411_OE_miR_150_5p_1
hsa-miR-450a-5p · A2780	GSE129076_OE_miR_450a_5p_1
hsa-miR-450a-5p · SKOV-3	GSE129076_OE_miR_450a_5p_2
hsa-miR-1271-5p · SH-SY5Y	GSE148827_OE_miR_1271_5p
hsa-miR-21-3p · HaCaT	GSE154968_OE_miR_21_3p
hsa-miR-20b-5p · HepG2	GSE168484_OE_miR_20b_5p
hsa-miR-100-3p · Kelly	GSE175729_OE_miR-100-3p_Kelly
hsa-miR-1915-3p · Kelly	GSE175729_OE_miR-1915-3p_Kelly
hsa-miR-380-3p · Kelly	GSE175729_OE_miR-380-3p_Kelly
hsa-miR-485-3p · Kelly	GSE175729_OE_miR-485-3p_Kelly
hsa-miR-99b-5p · Kelly	GSE175729_OE_miR-99b-5p_Kelly
hsa-miR-130a-3p · bone-marrow mesenchymal stem cells	GSE202135_OE_miR_130a_3p
hsa-miR-130b-3p · bone-marrow mesenchymal stem cells	GSE202135_OE_miR_130b_3p
hsa-miR-375-3p · HUV-EC-C	GSE210778_OE_miR_375_3p